

UBERABA, Brazil

WMO: 835760

Lat: 19.765S

Lon: 47.965W

Elev: 2655

StdP: 13.34

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	52.1	55.8	32.4	29.6	77.8	35.7	33.8	77.3	19.3	72.8	17.4	73.1	6.4	240	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	19.0	93.3	65.0	91.4	65.9	89.6	66.6	74.0	82.4	73.0	81.8	72.4	81.2	8.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	130.2	75.7	71.3	127.6	75.2	70.1	122.3	74.4	39.4	82.3	38.6	82.6	38.0	81.2	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.7	13.9	DB	46.1	95.7	4.7	2.4	42.7	97.4	39.9	98.9	37.3	100.2	33.9	102.0	(4)
(5)			WB	42.3	76.6	4.6	2.8	39.0	78.7	36.3	80.3	33.8	81.9	30.4	83.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.8	76.7	77.4	76.6	74.9	71.4	69.8	70.2	73.5	76.8	77.7	76.2	76.5		
	DBStd	4.56	2.82	2.34	2.59	3.35	3.86	4.12	4.43	4.00	4.69	4.49	3.31	2.59		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	41	0	0	0	1	6	13	16	4	1	0	0	0		
	CDD50	9050	827	768	824	748	662	594	627	727	803	860	787	823		
	CDD65	3617	362	348	359	299	203	157	178	266	355	395	337	358		
	CDH74	26862	2404	2501	2299	2035	1236	992	1274	2312	3469	3685	2367	2288		
	CDH80	9170	804	875	735	589	243	151	257	796	1557	1637	808	718		
(14) Wind	WSAvg	7.2	6.9	6.5	6.4	6.5	6.4	6.9	7.3	7.8	8.4	7.9	7.7	7.3		
(15) Precipitation	PrecAvg	58.4	11.4	8.7	7.8	2.8	1.9	0.7	0.4	0.6	2.2	4.2	7.4	10.0		
	PrecMax	71.8	22.4	13.6	14.6	6.1	4.8	3.3	2.5	2.3	5.1	9.5	14.1	14.3		
	PrecMin	43.0	3.8	4.1	4.2	0.0	0.1	0.0	0.0	0.0	0.1	1.7	4.3	5.5		
	PrecStd	7.6	4.7	3.2	2.7	1.6	1.3	0.9	0.6	0.6	1.4	2.0	2.5	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.1	91.5	89.9	89.9	86.4	85.6	86.3	91.7	95.1	97.0	92.8	89.9		
		MCWB	69.5	70.8	70.0	70.6	65.7	62.7	61.8	62.1	63.1	64.6	68.3	69.7		
	2%	DB	89.2	89.5	88.0	87.6	84.4	83.1	84.5	89.3	93.2	93.6	89.6	88.0		
		MCWB	69.4	69.9	70.0	69.2	65.6	62.9	61.6	61.6	63.5	65.8	68.4	69.8		
	5%	DB	87.4	87.7	86.2	85.7	82.6	82.1	82.7	86.8	91.2	91.4	87.6	86.2		
		MCWB	69.5	69.8	70.0	68.3	64.9	62.7	61.3	60.8	63.2	66.0	68.7	70.1		
	10%	DB	84.6	85.8	84.4	83.9	80.8	80.3	80.9	84.4	88.2	89.2	84.6	84.3		
		MCWB	70.2	69.9	70.1	68.1	64.5	62.5	60.9	60.4	63.1	66.2	68.7	70.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.7	74.9	73.8	73.9	70.8	68.3	67.3	66.8	71.1	72.8	74.5	75.2		
		MCDB	81.6	84.4	83.1	83.4	79.5	77.6	77.4	81.6	84.9	86.1	82.2	81.6		
	2%	WB	73.5	73.5	72.8	71.9	69.2	66.6	65.1	65.0	69.3	71.1	72.5	73.3		
		MCDB	80.9	82.9	82.6	80.9	77.7	76.9	77.3	79.3	81.6	82.5	81.1	81.3		
	5%	WB	72.5	72.5	72.0	71.0	68.1	65.5	63.7	63.7	68.0	70.0	71.4	72.3		
		MCDB	80.2	81.8	81.7	79.7	76.9	76.1	77.4	78.2	79.9	80.3	80.3	80.6		
	10%	WB	71.8	71.7	71.3	70.0	66.6	64.2	62.5	62.5	66.6	69.1	70.5	71.6		
		MCDB	79.5	80.8	80.5	78.4	75.6	74.9	76.2	77.7	78.5	79.5	78.8	80.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.6	16.0	17.2	17.8	19.3	20.3	21.5	21.0	19.0	16.7	15.8		
		MCDBR	18.9	19.2	18.4	19.1	20.0	21.4	22.2	23.5	23.3	22.1	20.1	18.5		
	5% WB	MCWBR	6.3	6.3	6.2	6.6	7.5	8.0	7.9	7.8	7.5	6.6	6.6	6.2		
		MCWBR	15.7	16.9	16.3	16.5	17.2	18.5	20.4	21.0	19.7	18.6	16.3	15.7		
(40) Clear-Sky Solar Irradiance	taub		0.384	0.383	0.371	0.356	0.336	0.335	0.344	0.373	0.469	0.439	0.395	0.390		
		taud	2.471	2.458	2.510	2.521	2.534	2.539	2.485	2.378	2.114	2.263	2.405	2.443		
	Ebn at Noon		305	301	297	290	284	278	279	279	264	281	299	303		
		Edn at Noon	38	38	35	32	29	28	30	36	50	45	40	39		
(44) All-Sky Solar Radiation	RadAvg		1791	1876	1702	1626	1422	1342	1429	1673	1766	1854	1782	1818		
(45)	RadStd		217	148	149	103	72	68	78	96	162	137	93	120		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page